

Gijs Joppe Moens

Curriculum Vitae

Amsterdam

Personal Website

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🌐 GijsMoens

in gijs-moens-051861123

Date of Birth: 17 July 1998 · Nationality: Dutch



Professional Experience

- Sep 2023 – Present **Researcher**, *Netherlands Cancer Institute*, Amsterdam
- Developed SONIC (Spectral Oriented Neural Invariant Convolutions), a continuous spectral approach to resolution-invariant image segmentation. First-author paper accepted at **ICLR 2026**.
 - Achieved performance matching or exceeding nnU-Net with an order of magnitude fewer parameters on 3D medical imaging benchmarks (KiTS, ACDC, AMOS).
- Jun 2021 – **Lead Researcher**, *Province of South Holland*, The Hague
- Apr 2026
- Lead a small AI R&D team; bridge organisational requirements with feasible ML solutions.
 - Architect and prototype a compliance analytics platform for integrity investigations.
 - Designed PUZZLE, a self-supervised spatial embedding method for geographic regions using redundancy reduction and proximity-based supervision on H3 grids.
- 2018 – 2020 **Compliance Officer**, *Various Trust Offices*, The Hague
- Ensured adherence to Dutch and EU financial regulations; drafted internal control protocols.

Education

- Sep 2022 – **M.Sc. Artificial Intelligence**, *University of Amsterdam*, Amsterdam
- Jul 2024
- GPA: 8.2/10
 - Focus areas: geometric deep learning, graph neural networks, reinforcement learning, signal processing.
- Sep 2019 – **B.Sc. Business Analytics**, *Vrije Universiteit Amsterdam*, Amsterdam
- Jul 2022
- GPA: 7.5/10
 - Interdisciplinary programme combining mathematics, computer science, and economics.
- Aug 2021 – **Exchange Minor**, *Hanken School of Economics*, Helsinki
- Dec 2021
- Advanced modules in mathematics and computer science.

Publications & Research

- Moens, G. J., Beets-Tan, R., & Pooch, E. H. P. (2026). SONIC: Spectral Oriented Neural Invariant Convolutions. *ICLR 2026*. arXiv · OpenReview
- Moens, G. J., de Witte, J., Göbel, T. P., & van den Oever, M. A. A. (2023). Reproducibility study of fairness-enhanced node representation learning. *ReScience C*, 9. OpenReview
- Scalar representation on equivariant graph networks for molecular force prediction. Code
- Enhancing Monte Carlo Tree Search under uncertainty. Code
- Genetic algorithm for the multi-trip multi-vehicle routing problem. Code
- Full-stack development of a French homestay booking site. Website

Technical Skills

Programming: Python (advanced), C++, CSS, HTML5, JavaScript, SQL (working proficiency)

Frameworks: PyTorch, JAX, CUDA, NumPy, Docker, AWS, Git, Neovim

Languages: Dutch (native), English (fluent)